

Thiovibrio frasassiensis gen. nov., sp. nov., an autotrophic, elemental sulphur disproportionating bacterium isolated from sulphidic karst sediment, and proposal of *Thiovibrionaceae* fam. nov.

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Abstract

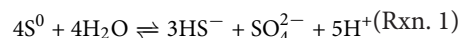
A novel, autotrophic, mesophilic bacterium, strain RS19-109^T, was isolated from sulphidic stream sediments in the Frasassi Caves, Italy. The cells of this strain grew chemolithoautotrophically under anaerobic conditions while disproportionating elemental sulphur (S⁰) and thiosulphate, but not sulphite with bicarbonate/CO₂ as a carbon source. Autotrophic growth was also observed with molecular hydrogen as an electron donor, and S⁰, sulphate, thiosulphate, nitrate and ferric iron as electron acceptors. Oxygen was not used as an electron acceptor and sulphide was not used as an electron donor. Weak growth was observed with sulphate as an electron acceptor and organic carbon as an electron donor and carbon source. The strain also showed weak growth by fermentation of tryptone. It grew at pH 5.5–7.5 (optimum, pH 7.0), 4–35 °C (optimum, 30 °C) and between 0–1.7% NaCl. Strain RS19-109^T was found to be phylogenetically distinct based on 16S rRNA gene sequence similarity (89.2%) to its closest relative, *Desulfurivibrio alkaliphilus* AHT2^T. The draft genome sequence for strain RS19-109^T had average nucleotide identity, average amino acid identity and *in silico* DNA–DNA hybridization values of 72.2, 63.0 and 18.3%, respectively, compared with the genome sequence of *D. alkaliphilus* AHT2^T. On the basis of its physiological and genomic properties, strain RS19-109^T is proposed as the type strain of a novel species of a novel genus, *Thiovibrio frasassiensis* gen. nov., sp. nov. A novel family, *Thiovibrionaceae* fam. nov., is proposed to accommodate *Thiovibrio* within the order *Desulfobulbales*. Strain RS19-109^T has been deposited at the DSMZ-German Collection of Microorganisms and Cell Cultures (=DSM 115074^T) and the American Type Culture Collection (=ATCC TSD-325^T).

DATA SUMMARY

Supplementary files available in the online version of this article can be found at Figshare 10.6084/m9.figshare.22665817 [1].

INTRODUCTION

Sulphur disproportionation, or the simultaneous oxidation and reduction of intermediate oxidation state sulphur compounds commonly yielding sulphate and sulphide (Rxn. 1), is a relatively understudied branch of the sulphur cycle compared to sulphate reduction and sulphide oxidation.



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Keywords: disproportionation; karst; sediment; sulphur.

Abbreviation: S⁰, elemental sulfur.

The GenBank accession number for the 16S rRNA gene sequence of strain RS19-109^T is OP597810. The GenBank accession number for the draft genome sequence of strain RS19-109^T is JAPHEH000000000.

Supplementary material is available with the online version of this article at Figshare 10.6084/m9.figshare.22665817.

Most micro-organisms capable of sulphur disproportionation are able to use thiosulphate and sulphite, but the ability to disproportionate elemental sulphur (S^0) is less common [2]. To date, 20 S^0 -disproportionating bacteria within the phyla *Desulfobacterota*, *Proteobacteria*, *Firmicutes* and *Nitrospirae* have been isolated from freshwater sediments [3–5], marine sediments [6–9], freshwater and soda lakes [10–12], hydrothermal vents [13–16], acidic river sediments [17], and hot springs [18–20]. In this study, we report the isolation and characterization of a novel S^0 -disproportionating bacterium, strain RS19-109^T, from sulphidic stream sediments in the karst at Frasassi, Italy. This is the first S^0 -disproportionating micro-organism isolated from the terrestrial subsurface. Based on its physiological features and genomic sequence, this strain represents a novel genus and species for which we propose the name *Thiovibrio frasassiensis* gen. nov., sp. nov. We also propose that this strain is a member of a novel family which we designated *Thiovibrionaceae* fam. nov.

METHODS

Enrichment and isolation

Strain RS19-109^T was isolated from a sediment sample collected from a shallow sulphidic stream in the Frasassi Caves, Italy (43.4012°N 12.9656°E) in July 2019. The sampling point was located in the Ramo Sulfureo of the Grotta del Fiume at 400 m below the surface [21]. Water temperature and pH at the sampling site were 13 °C and pH 7.3, respectively. Sediment samples were collected using a syringe, transferred into serum bottles purged with nitrogen and sealed with butyl stoppers, and transported back to the laboratory at the Geological Observatory at Coldigioco, Italy.

For enrichment of sulphur disproportionators, a bicarbonate-buffered freshwater minimal medium was prepared under 80% N_2 –20% CO_2 . The medium was based on DSMZ 386 but acetate was omitted, amorphous ferric hydroxide was added as a sulphide scavenger and sodium pyrosulphite was replaced with 10 g l⁻¹ S^0 . The basal medium contained the following (per litre of deionized water): 0.20 g KH_2PO_4 , 0.25 g NH_4Cl , 1.0 g NaCl, 0.40 g $MgCl_2 \cdot 6 H_2O$, 0.50 g KCl, 0.15 g $CaCl_2 \cdot 2 H_2O$, 1 ml trace element solution SL-10 (see DSMZ medium 320), 1 ml selenite–tungstate solution (see DSMZ medium 385), and 1 ml DSMZ seven vitamin mixture solution (see DSMZ medium 503). Amorphous ferric hydroxide was prepared as described by Lovley and Phillips [22] and was added to the medium at a final concentration of 30 mM [22]. The pH of the medium was adjusted to pH 7.0 using NaOH. Medium was dispensed into serum bottles containing 10 g l⁻¹ S^0 and was sterilized by heating for 2–3 h in a boiling water bath on each of three successive days. After sterilization, sodium bicarbonate was added at a final concentration of 30 mM and sodium sulphide was added as a reductant (final concentration 0.50 μ M). The complete medium was inoculated with Frasassi sediment (1% w/v).

Measurements of dissolved and adsorbed ferrous iron using a modified version of the ferrozine assay [23] were used as partial proxies for sulphide concentration. Samples were acidified in 0.5 M HCl, further diluted with 1 M HCl, mixed with ferrozine solution, and absorbance was measured at 562 nm. The addition of hydroxylamine hydrochloride, which reduces Fe(III) to Fe(II) was omitted. Sulphate concentrations were measured using a turbidimetric sulphate assay [24]. After 1 year of incubation at 15 °C, the culture was transferred and purified by dilution-to-extinction. Purity of the resulting isolate was confirmed in the second transfer by microscopy and Sanger sequencing of the 16S rRNA gene, and the resulting isolate was designated as strain RS19-109^T.

Light and electron microscopy

Growth of strain RS19-109^T was determined by direct counts using fluorescence microscopy. Culture (1 ml) was fixed in glutaraldehyde (2.5% final concentration) and stained with 125 μ l 10 $\times$ SYBR Green I. Cultures with ferric iron were treated with an oxalate solution (28 g l⁻¹ ammonium oxalate and 15 g l⁻¹ anhydrous oxalic acid) prior to staining to remove iron particles [25]. Oxalate solution (400 μ l) was added to 1 ml of culture and was incubated for 15 min at room temperature, neutralized with NaOH, and stained with 125 μ l 10 $\times$ SYBR Green I. Samples were filtered through black polycarbonate filters with a 0.22 μ m pore size and cells were counted under a fluorescence microscope.

Scanning electron microscopy was used to obtain images of strain RS19-109^T cells. Cells were fixed in 2% glutaraldehyde for 1 h at room temperature, centrifuged for 2 min at 10000 r.p.m. to pellet the minerals, and the supernatant was filtered onto a 0.1 μ m Supor filter. Samples were dehydrated in an ethanol series (30, 50, 70, 80, 90, 95 and 100%), dried using a Tousimis 815 critical point dryer, and sputter-coated for 40 s with Pt/Pd using a Cressington 108 manual sputter coater. Cells were observed with a Nova NanoSEM 450 field emission scanning electron microscope at an accelerating voltage of 5.0 kV.

Temperature, pH, and salt tolerance

Growth experiments to evaluate tolerance of temperature, pH and salinity were performed in triplicate in serum bottles with the same basal medium used for pure culture isolation. The temperature range for growth was evaluated at 4–50 °C at 5 °C intervals at pH 7.0. The pH range for growth was evaluated at 30 °C over a range of pH 5.0 to 8.5 at intervals of 0.5 pH unit. The culture medium was supplemented with the following buffers at final concentrations of 25 mM: MES (pH 5.0–6.5)

and HEPES (pH 7.0–8.5). Salinity tolerance of strain RS19-09^T was evaluated at 30 °C using NaCl-free basal growth medium supplemented with NaCl (0, 0.1, 0.25, 0.5, 0.75, 1, 1.2, 1.5, 1.7, 2.0%).

Physiology

All phenotypic tests were performed in triplicate in serum bottles. Chemolithoautotrophic growth on S⁰ as a substrate for disproportionation was tested in the basal medium with 10 g l⁻¹ S⁰ added as the electron donor and acceptor, bicarbonate/CO₂ added as the carbon source, and amorphous ferric oxyhydroxide as a sulphide scavenger. Disproportionation of thiosulphate (20 mM) and sulphite (5 mM) in place of S⁰ were tested in the basal medium. Reduction of sulphate (20 mM) was tested using bicarbonate-free basal medium with the addition of 10 mM of each of the following substrates to the medium: acetate, citrate, ethanol, formate, fumarate, glucose, lactate, pyruvate, succinate, tryptone and yeast extract (0.1%). The growth tests were conducted with and without the addition 0.001% yeast extract. The use of carbon source as an electron donor for sulphate reduction or as a substrate for fermentation was determined by measuring sulphide with the Cline methylene blue assay [26]. Production of sulphide indicated that the carbon source was used as an electron donor for sulphate reduction, while growth without concomitant sulphide production indicated fermentative growth. Electron acceptor growth tests were performed with 20 mM of S⁰, sulphate, thiosulphate, sulphite, nitrate or amorphous ferric hydroxide (with the exception of sulphite – 5 mM) in the basal medium with N₂/CO₂ replaced by 80% H₂–20% CO₂. Aerobic respiration was tested with S⁰ and sulphide as electron donors in the basal medium. Oxygen was added to the headspace by placing a needle fitted with a syringe filter through the butyl rubber stopper.

16S rRNA phylogeny and genome features

Genomic DNA was extracted from strain RS19-109^T using the Qiagen DNeasy UltraClean Microbial Kit and was used for 16S rRNA gene sequencing and whole genome sequencing. The 16S rRNA gene was amplified using 27F/1492R primers and was sequenced by GeneWiz (South Plainfield, NJ). 16S rRNA gene sequences from close cultivated relatives identified using BLAST from the NCBI ref_seq_rna database were aligned using MUSCLE version 3.8.31 [27]. Gaps were removed and sequences were trimmed using trimAl version 1.2 [28]. A phylogenetic tree was reconstructed using the aligned and trimmed 16S rRNA gene sequences using IQ-TREE with a maximum-likelihood method and the generalized time-reversible model of nucleotide evolution with 1000 bootstrap replications [29]. To obtain 16S rRNA gene sequence identities, sequences were aligned using the SILVA SINA aligner version 1.2.11 [30, 31], gaps were removed, and identities were calculated in Geneious v6.1.8 (www.geneious.com).

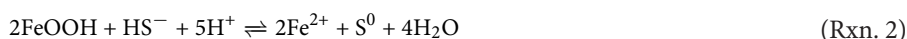
The genome of strain RS19-109^T was sequenced by Novogene (Sacramento, CA) on an Illumina NovaSeq platform yielding 7079420 paired-end reads. The sequence quality of the reads was determined with FastQC version 0.11.8 [32]. The read length was 150 bp. Error-corrected reads were assembled with SPAdes version 3.15.4 [33]. Genome quality was determined using CheckM [34]. Annotation of the RS19-109^T draft genome was performed using MetaSanity [35] and METABOLIC [36] (Supplementary file 3).

Closely related strains were identified using GToTree [37]. The 'gtt-get-accessions-from-GTDB' command was used to collect representative genomes from the order *Desulfobulbales* present in the Genome Taxonomy Database [38]. Pairwise average nucleotide identity (ANI), average amino acid identity (AAI) and digital DNA–DNA hybridization (dDDH) values were calculated for all available genomes in the *Desulfobulbales* using the IMG ANIcalculator [39], EzAAI [40] and the Genome-to-Genome Distance Calculator version 3.0 [41, 42], respectively. Genomes from isolated species and genomes from the closest uncultured relatives of strain RS19-109^T were used to reconstruct a phylogenomic tree (Fig. 4). GTDB-tk (version 2.1.1) was used to align 120 conserved single-copy marker gene sequences (bac120) from the genomes [38, 43, 44]. IQ-TREE was used to reconstruct a tree from the aligned sequences using the methods described above [29].

RESULTS AND DISCUSSION

Morphological and physiological characteristics

After isolation, the culture incubated for 40 days at 15 °C. Cell densities increased from 10⁴ cells ml⁻¹ to 10⁸ cells ml⁻¹ (Fig. 1c) and the medium colour changed from rusty red to black, indicating reduction of ferric iron to ferrous iron by the reaction with sulphide (Rxn. 2) and the formation of iron sulphide (Rxn. 3).



After 40 days, average sulphate concentrations increased to over 21.4 mM, average ferrous iron concentrations increased to 1.8 mM (Fig. 1a, b) and all of the red amorphous ferric hydroxide (30 mM) was converted to black precipitates. The black precipitates that formed were likely composed of both acid-volatile sulphides (AVS; H₂S and FeS), which can be measured by acidification in the Ferrozine assay, and chromium (II)-reducible sulphides (CRS; S⁰, pyrite, FeS₂) [45, 46]. Pyrite can be formed from the reaction of FeS with S⁰ or polysulphides [47, 48]

The net reaction of S⁰ disproportionation and FeS production can be written as:

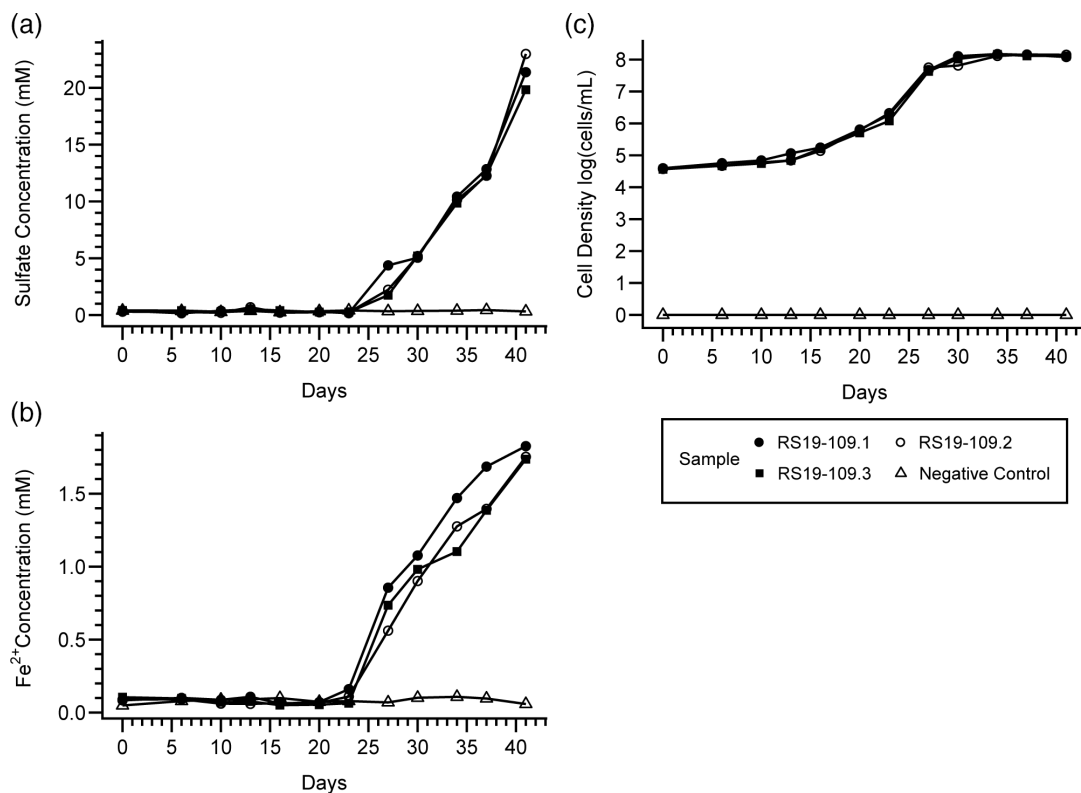


Fig. 1. Changes in (a) sulphate concentration (mM), (b) ferrous iron concentration (mM), and (c) cell density (log(cells/ml)) as a function of time (days) of three RS19-109^T cultures grown at 15°C in basal S⁰ disproportionation medium amended with ferric iron. Open triangles represent negative, uninoculated control bottles.



Assuming that 30 mM of total sulphide was produced, the ratio of sulphide to sulphate that was observed was 1.4, which is about 30% less than what would be expected by the stoichiometry of Rxn. 4. A similar discrepancy (1.54:1) was observed by [45], who attributed it to difficulties in sampling acid-resistant grey coatings on sulphur particles. Future studies of strain RS19-109^T should include measurements of both AVS and CRS to determine more accurate stoichiometric ratios. Although we were not able to perform these measurements in this study, the production of both sulphide and sulphate in our cultures can be explained by S⁰ disproportionation.

Cells of strain RS19-109^T were vibrio-shaped bacteria and occur singly, 0.6 µm in diameter and 2.3 µm long (Fig. 2). The temperature optimum was 30°C. No growth was detected above 36°C after incubation for 4 weeks. The optimum pH was pH 7.0 and no growth was observed below pH 5.5 or above pH 7.5. The salinity optimum was 0–1.7% NaCl. No growth was observed above 1.7% NaCl.

Strain RS19-109^T was able to disproportionate S⁰ and thiosulphate, but not sulphite. Growth by S⁰ or thiosulphate disproportionation was accompanied by production of sulphate and conversion of amorphous ferric hydroxide to a black precipitate, indicating the production of ferrous iron by the reaction of sulphide with ferric iron. Strain RS19-109^T showed weak growth as a sulphate reducer using acetate, ethanol, formate, fumarate, glucose, pyruvate and yeast extract (0.1%). It did not reduce sulphate with tryptone, but instead fermented tryptone leading to weak growth without the production of sulphide. No growth was observed with citrate, lactate or succinate. Strain RS19-109^T used H₂ as an electron donor with S⁰, sulphate, thiosulphate, nitrate and amorphous ferric hydroxide as electron acceptors. It was also able to reduce ferric iron with acetate (10 mM). Sulphite was not used as an electron acceptor. Acetate was not used as an electron donor for nitrate reduction, with or without the addition of 0.001% yeast extract. No growth was observed with oxygen as the electron acceptor. Differential physiological and biochemical characteristics that distinguished strain RS19-109^T and close relatives *Desulfurivibrio alkaliphilus* AHT2^T, *Desulfobulbus propionicus* 1pr3^T, *Desulfocapsa thiozymogenes* Bra2^T and *Dissulfuribacter thermophilus* S69^T are listed in Table 1.

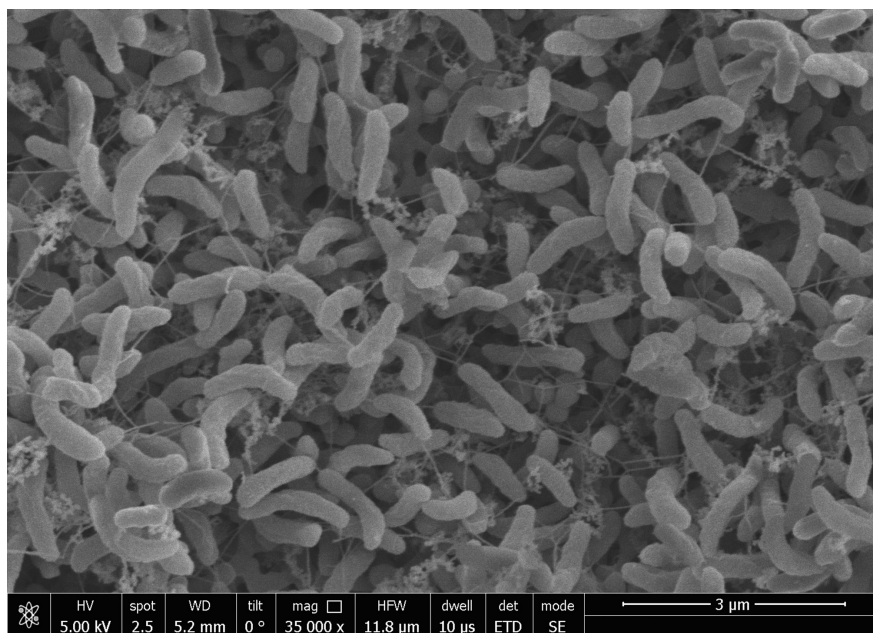


Fig. 2. Scanning electron microcopy image of cells of strain RS19-109^T grown on basal S⁰ disproportionation medium containing Fe(III) as a sulphide scavenger.

Phylogenetic and genomic characteristics

The draft genome sequence of strain RS19-109^T contains 2841614 bp in three contigs and was 98.13% complete and 0.6% contaminated. The G+C content of the genomic DNA was 56.25 mol%, the N50 was 2800286 and the sequence depth of coverage was 374×. A total of 2621 protein-coding sequences and 48 tRNA-encoding genes were predicted. The draft genome of strain RS19-109^T contained one copy each of 5S, 16S and 23S rRNA genes.

The closest recognized species to strain RS19-109^T based on 16S rRNA gene sequence similarity was *Desulfurivibrio alkaliphilus* AHT2^T (89.2%) (Fig. 3) [11], which is far below the threshold of 98.65% used to differentiate species [49, 50] and the 94.5% sequence similarity used to differentiate genera [51]. Using the median value for the rank of family designated by Yarza and colleagues (<92.25% [51]) and the range for novel families designated by Konstantinidis and colleagues (89–92% [50]), we propose that strain RS19-109^T belongs to a novel species, genus, and family. In a phylogenomic tree, strain RS19-109^T and its closest uncultured relatives formed a clade that clustered most closely with the family *Desulfurivibrionaceae*, outside of the families *Desulfobulbaceae* and *Desulfocapsaceae* (Fig. 4) [52]. Genomes from the closest uncultured relatives, designated by UBA2262, were recovered from wastewater from the Athabasca oil sands in Canada [53]. For genomic discreteness, the ANI value (72.2%) and dDDH value (18.3%) of strain RS19-109^T fall well below the species-level ANI cutoff and the dDDH cutoff for relatedness to its closest cultured relative (<95% and <70%, respectively) (Table 2, supplemental files 1 and 2) [50]. The proposed AAI thresholds for designating genera and families are 65–95% and 45–65%, respectively [50]. When compared to the type genera of the families within the *Desulfobulbales* designated by Waite *et al.* [52], strain RS19-109^T has AAI values of 58.6–63%, which would classify it as a novel genus within the family *Desulfurivibrionaceae*. However, when these type genera are compared to each other, *Desulfurivibrio alkaliphilus* AHT2^T, *Desulfobulbus propionicus* 1pr3^T and *Desulfocapsa thiozymogenes* Bra2^T all had AAI values to each other and to RS19-109^T of at least 58%, putting them all well above the 45% AAI family level cutoff that would designate them as novel families (Table 2 [50]). On the basis of its phylogenomic placement and 16S rRNA gene sequence similarity to close relatives, we propose that strain RS19-109^T is placed within a novel family in the order *Desulfobulbales*, which we designate *Thiovibrionaceae*.

The genome of RS19-109^T contained genes encoding enzymes involved in carbon fixation via the Wood–Ljungdahl pathway (acetyl-CoA synthase *cdhDE/acs*; carbon monoxide dehydrogenase *cooS*). Strain RS19-109^T also has the genetic potential for acetogenesis (acetyl coA synthetase *acdA*; acetate kinase *ack*; phosphate acetyl transferase *pta*), formate oxidation (formate dehydrogenase *fdoH*) and pyruvate oxidation (pyruvate ferredoxin oxidoreductase *porA*). The genome contained genes encoding nitrogenases (*nifDKH*) and a dissimilatory nitrate reduction pathway (nitrate reductase *napAB*; nitrate reductase *nrfAH*). Genes for flagellum biosynthesis were present in the genome, suggesting motility via flagellum.

Table 1. Characteristics that distinguish strain RS19-109^T from representatives of closely related species

Strains: 1, RS19-109^T (data from this study); 2, *Desulfurivibrio alkaliphilus* AHT2^T (=DSM 19089^T; data from [11]); 3, *Desulfobulbus propionicus* 1pr3^T (=DSM 2032^T; data from [3, 4]); 4, *Desulfocapsa thiozymogenes* Bra2^T (=DSM 7269^T; data from [5]); 5, *Dissulfuribacter thermophilus* S69^T (=DSM 25762^T; data from [14]). NR, Not reported; +, positive; –, negative.

Characteristic	1	2	3	4	5
16S rRNA gene sequence similarity (%) to RS19-109 ^T	100	87.4	80.0	83.9	
Source of isolation	Sulphidic karst sediment	Haloalkaline lake sediment	Freshwater sediment	Freshwater sediment	Hydrothermal vent chimney
Morphology	<i>Vibrio</i>	<i>Vibrio</i>	Ellipsoidal/lemon-shaped	Rod	Rod
Temperature range for growth (°C)	4–36	Up to 42 °C	10–43	20–30	28–70
Optimum temperature for growth (°C)	30	Mesophilic (NR)	39	NR	61
pH range for growth	5.5–7.5	8.5–10.3	6.0–8.6	6.8–8.1	5.6–7.9
Optimum pH for growth	7.0	9.5	7.2	7.3–7.5	6.8
NaCl range for growth (% w/v)	0–1.7	1.17–14.61	0–1.5	0–1.5	0.9–5.0
Optimum NaCl for growth (% w/v)	0–0.1	2.92–5.84	NR	NR	1.8–2.7
Electron acceptors used:					
S ⁰	+	+	–	NR	+
Sulphate	+	–	–	+	–
Thiosulphate	+	+	+	–	+
Sulphite	–	–	+	–	NR
Fe(III)	+	NR	NR	–	+
Nitrate	+	+	+	–	–
Oxygen	–	NR	NR	NR	–
Electron donors used:					
Hydrogen	+	+	+, in presence of acetate and CO ₂	–	+
Organic C	Acetate, ethanol, formate, fumarate, glucose, pyruvate, yeast extract	Formate, ethanol, pyruvate, lactate, fumarate, glycerol, butyrate, isobutyrate, malate, succinate, yeast extract	Propionate, lactate, pyruvate, ethanol, propanol	Ethanol, propanol, butanol	Fumarate, maleinate
Sulphide	–	+	NR	NR	NR
Fermentation	Tryptone	NR	Lactate	–	–
Disproportionation:					
S ⁰	+	+	+	+	+
Thiosulphate	+	NR	NR	+	+
Sulphite	–	NR	NR	+	+
DNA G+C content (mol%)	56.25*	60.3*	59.9	50.7	40.5

*Value determined from genome sequence.

Several genes encoding enzymes in sulphur metabolism were present in the genome of RS19-109^T, including a complete dissimilatory sulphate reduction pathway (sulphate adenylyltransferase *sat*; adenylylsulphate reductase *aprAB*; dissimilatory sulphite reductase *dsrAB*; other dissimilatory sulphite reductase subunit genes *dsrCDJKMOPT*), sulphur oxidation (sulphur dioxygenase *sor*) and thiosulphate disproportionation (thiosulphate reductase/polysulphide reductase *phsA*). The genome also contained genes for arsenate reduction (arsenate reductase *arsC2*) and iron/manganese reduction (decaheme-associated outer membrane protein *mtrB*). Genes for hydrogen oxidation (Ni-Fe hydrogenase *nife*-group 1), cytochrome bd ubiquinol oxidase (*cydAB*), NADH-quinone oxidoreductase Complex I (*nuoAB*) and F-type H⁺ transporting ATP-ase (*atpAD*) were also

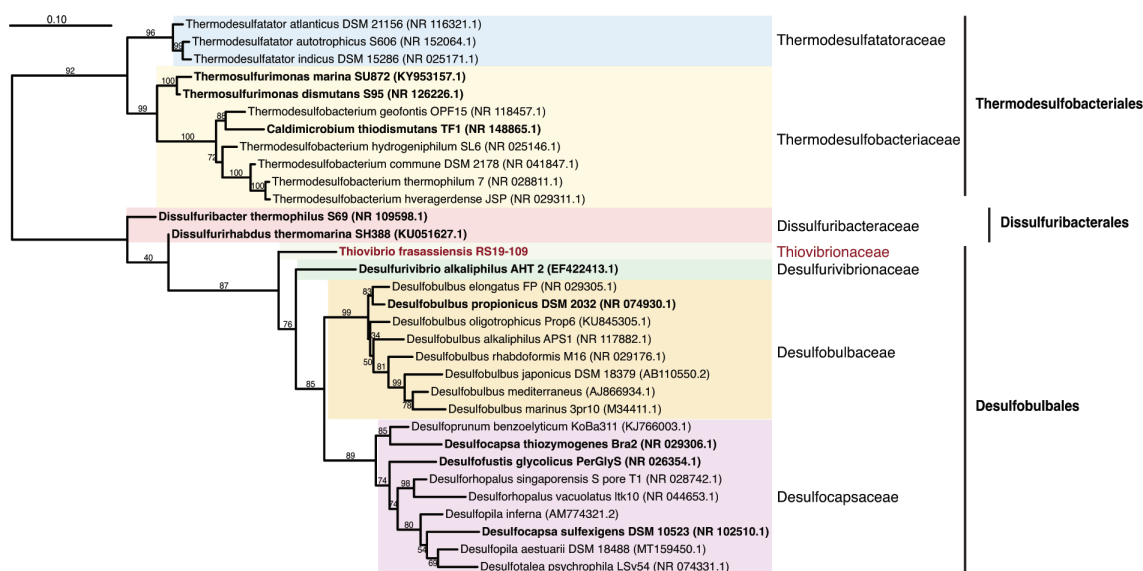


Fig. 3. Maximum-likelihood phylogenetic tree of 16S rRNA gene sequences showing the phylogenetic position of strain RS19-109^T. Bootstrap values (%) are based on 1000 replicates. NCBI accession numbers for 16S rRNA gene sequences are shown in parentheses. Bar represents 0.1 substitutions per nucleotide position. Strain RS19-109^T is 89.2% similar to its closest relative, *Desulfurivibrio alkaliphilus* AHT2^T, by 16S rRNA gene sequence similarity.

present. The genes and pathways present in the genome are consistent with the ability of strain RS19-109^T to reduce sulphate with hydrogen, disproportionate thiosulphate, reduce iron, reduce nitrate and utilize bicarbonate/CO₂ as a carbon source. To date, genomic markers for S⁰ disproportionation have not been identified [2, 54] and we were thus unable to ascribe any genes or pathways to the ability of RS19-109^T to disproportionate S⁰.

DESCRIPTION OF *THIOVIBRIO* GEN. NOV. NON *THIOVIBRIO* JANKE 1924 [55] NON *THIOVIBRIO* EMOTO AND HIROSE 1942 [56]

Thiovibrio (Thi.o.vi'bri.o., Gr. neut. n. *theion*, sulphur; N.L. masc. n. *vibrio*, a vibrio; N.L. masc. n. *Thiovibrio*, a sulphur vibrio).

Cells are vibrio shaped. Anaerobic, mesophilic and neutrophilic. Chemolithoautotrophic growth by disproportionation of S⁰ to sulphide and sulphate. Member of the phylum *Desulfobacterota*. The type species is *Thiovibrio frassasiensis*.

DESCRIPTION OF *THIOVIBRIO FRASSASIENSIS* SP. NOV.

Thiovibrio frassasiensis (fra.sas.si.en'sis. N.L. masc. adj. *frassasiensis* pertaining to Frasassi, the location of the caves from which the type strain was isolated).

Shows the following properties in addition to those given in the genus description. Cells are vibrio shaped and occur singly, 0.6 µm in diameter and 2.3 µm long. The temperature range for growth is 4–36 °C, with an optimum at 30 °C. The pH range for growth is pH 5.5–7.5, with an optimum at pH 7.0. Grows in 0–1.7% (w/v) NaCl with an optimum at 0–0.1% (w/v). S⁰ and thiosulphate are disproportionated to form sulphate and sulphide with bicarbonate/CO₂ as a carbon source. Growth is enhanced in the presence of amorphous ferric hydroxide. In the presence of 80% H₂–20% N₂, S⁰, sulphate, thiosulphate, nitrate and amorphous ferric hydroxide serve as electron acceptors. Weak anaerobic heterotrophic growth is observed on acetate, ethanol, formate, fumarate, glucose, pyruvate and yeast extract with sulphate as the electron acceptor. Fermentative growth is observed on tryptone. Does not oxidize or ferment citrate, lactate or succinate. Grows by the oxidation of H₂ with S⁰, sulphate, thiosulphate, nitrate and ferric iron. Does not grow by oxidizing sulphide or S⁰ with oxygen and does not use sulphite as an electron acceptor.

The type strain is RS19-109^T, isolated from sulphidic stream sediment sampled from the Grotta del Fiume in the Frasassi Gorge, Italy. The G+C content of the genomic DNA of the type strain is 56.25 mol% (determined from the genome sequence). Strain RS19-109^T has been deposited at the DSMZ-German Collection of Microorganisms and Cell Cultures (=DSM 115074^T) and the American Type Culture Collection (=ATCC TSD-325^T). The GenBank accession numbers for the 16S rRNA gene and draft genome sequences are OP597810 and JAPHEH000000000, respectively. Draft genome annotations are available with the supplementary material, available in the online version of this article at 10.6084/m9.figshare.22665817.

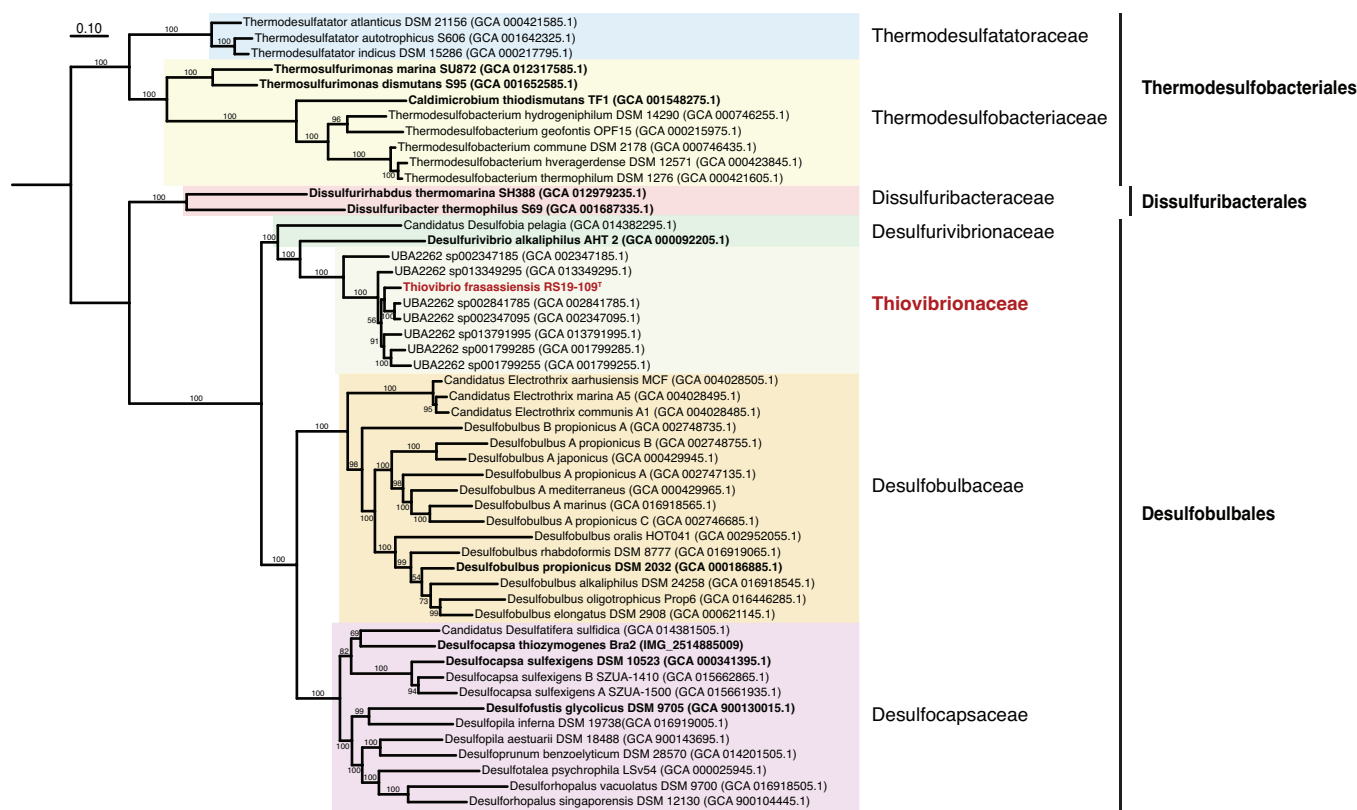


Fig. 4. Maximum-likelihood phylogenomic tree of 120 conserved single-copy marker gene sequences from strain RS19-109^T and 50 representative genomes of isolated and candidatus members of the orders *Desulfurivibrionales*, *Thermodesulfobacteriales* and *Dissulfuribacteriales*. Bootstrap values (%) are based on 1000 replicates. NCBI accession numbers for genome assemblies are shown in parentheses. Bar represents 0.1 substitutions per nucleotide position. Genomes designated 'UBA2262' are the closest uncultured relatives of strain RS19-109^T.

Table 2. Pairwise comparisons of 16S rRNA gene sequence identities (%; values to the right of the grey boxes) and ANI/AAI (%; values to the left of the grey boxes) between strain RS19-109^T and type genera of the families *Desulfurivibrionaceae*, *Desulfobulbaceae* and *Desulfocapsaceae*

	RS19-109 ^T	<i>Desulfurivibrio alkaliphilus</i> AHT2 ^T	<i>Desulfobulbus propionicus</i> 1pr3 ^T	<i>Desulfocapsa thiozymogenes</i> Bra2 ^T
RS19-109 ^T	–	89.2	87.6	87.3
<i>Desulfurivibrio alkaliphilus</i> AHT 2 ^T	72.2/63.0	–	88.8	88.9
<i>Desulfobulbus propionicus</i> 1pr3 ^T	69.9/58.6	70.0/57.7	–	87.2
<i>Desulfocapsa thiozymogenes</i> Bra2 ^T	70.8/60.9	69.9/58.3	70.0/60.5	–

DESCRIPTION OF THIOVIBRIONACEAE FAM. NOV.

Thiovibrionaceae (Thi.o.vi.bri.o.na.ce'ae, N.L. masc. n. *Thiovibrio* type genus of the family; -aceae ending to denote family; N.L. fem pl. n. *Thiovibrionaceae*, the *Thiovibrio* family).

Encompasses chemolithoautotrophic bacteria. Based on the results of 16S rRNA gene sequence and whole genome analysis, the strain is phylogenetically affiliated with the order *Desulfobulbales*. The type genus is *Thiovibrio*.

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Conflicts of interest

The authors declare that there are no conflicts of interest.

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